

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I A Choose the correct answer for the following questions.

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1. Replication of circular DNA through Θ -model is

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|-------------------------|---------------------------------|
| (i) Unidirectional | (ii) Bidirectional |
| (iii) Both (i) and (ii) | (iv) Bidirectional needs primer |

2. If a genetic code is comma-less, it means that

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|---|--|
| (i) A given base triplet can code for more than one amino acids | (ii) There is no punctuation in code sequence |
| (iii) The third base is not important in coding | (iv) A given amino acid can be coded by more than one base triplet |

3. t-RNA specifies amino acids, during proteins synthesis. t-RNA is charged with amino acid at 3' end and its anticodon pairs with codon of m-RNA. Select the incorrect statement out of the following

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| (i) Information is transmitted between 3' end of t-RNA and anticodon |
| (ii) Codon-anticodon interaction occurs at A-site of ribosomes |
| (iii) Each t-RNA has specific anticodon |
| (iv) Each codon is triplet |

4. The enzyme peptidyl transferase facilitates

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| (i) Joining of two neighboring amino acids |
| (ii) Transfer of t-RNA between A to P site |
| (iii) Both (i) and (ii) |
| (iv) Formation of un-specified amino acids |

5. **Mushroom derived α -aminoglycosides inhibits**
 (i) RNA polymerase (ii) 70S Ribosomes (iii) 80S ribosomes (iv) m-RNA synthesis
6. **In prokaryotes, the DNA is replicated by**
 (i) 3'-5' exonuclease (ii) DNA Ligase (iii) DNA polymerase I (iv) Pol III
7. **What is the function of positively charged metal ions at active site of DNA polymerase?**
 (i) to bind ATP
 (ii) to stabilize the negative charge of the transition state
 (iii) to complex with a Tyr residues in the active site
 (iv) both (i) and (iii)
8. **Which of the following statements are correct? Please select all that apply**
 (i) RNA polymerase can synthesize new strands without primer
 (ii) DNA polymerase can synthesize new strand without primer.
 (iii) Telomerase does not need any primer.
 (iv) DNA ligase needs primer.
9. **The non-coding sequences of DNA is known as**
 (i) pseudogenes (ii) introns (iii) repetitive sequences (iv) all of the above
10. **Which of the following is true for topoisomerase I:**
 (i) change in linking no in the steps of two
 (ii) Responsible for maintaining proper DNA structure
 (iii) Make transient ds breaks in the DNA
 (iv) does not require ATP energy for its activity
11. **In nuclear splice sites the intron-exon junctions are often described as conforming to the**
 (i) GU-AG rule (ii) UG.....AG rule (iii) GU..... GA rule (iv) AG.....GU rule
12. **The consensus TTGACA denotes which element in a bacteria promoter?**
 (i) UP element (ii) -10 element (iii) -35 element (iv) none of the above
13. **Downstream promoter element is the essential feature of which of the following promoters?**
 (i) Prokaryotic pol II (ii) Prokaryotic pol I
 (iii) eukaryotic pol II (iv) none of above
14. **Attenuation is a mechanism observed in the regulation of which of the following operons?**
 (i) lac (ii) ara (iii) mer (iv) trp
15. **siRNAs inhibit expression of homologous genes in the following way(s):**
 (i) Trigger destruction of its mRNA
 (ii) Inhibit translation of its mRNA
 (iii) Induce chromatin modifications within the promoter that silence the gene
 (iv) All of the above

Q-IB	Match the items of Column A with appropriate item of Column B			5
	A		B	Answer
1.	Ribosome	A	Phage lambda	
2.	Si-RNA	B	Alternative structures	
3.	recA	C	Translation	
4.	cro	D	Homologous recombination	
5.	Triplex DNA	E	Non-coding	

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

- Q-IIA Answer the following questions as directed [Any Two] 8**
- 1 Provide experimental evidence for the hypothesis – DNA as genetic material
 - 2 Write a note on chemistry of DNA
 - 3 What is C-value? Discuss C-value paradox with suitable examples
- Q-IIB Differentiate the following[Any one] 4**
- 1 Topoisomerase I and Topoisomerase III
 - 2 Prokaryotic replication initiation and eukaryotic replication initiation
- Q-IIC Present a mechanism for the following through pictorial diagram [Any Two] 8**
- 1 Replication in retrovirus
 - 2 Homologous recombination
 - 3 Role of telomerase in solving end replication problem

SECTION – II

- Q-IIIA Answer any two of the following 10**
- 1 Differentiate prokaryotic and eukaryotic transcription initiation through pictorial diagram
 - 2 What is splicing? Discuss the mechanism for m-RNA splicing
 - 3 Write a note on eukaryotic transcription factors
- Q-IIIB Write a note on any two of the following: 10**
- 1 Eukaryotic translation factors
 - 2 t-RNA charging
 - 3 Post-translational processing of proteins
- Q-IIIC Answer any one of the following: 5**
- 1 Briefly discuss the events that take place in lactose utilizing bacteria in absence of lactose
 - 2 What is operon? Discuss the concept of tryptophan operon
- Q-IIID Answer any one of the following: 5**
- 1 What is chromatin remodeling? Discuss the mechanism of chromatin remodeling
 - 2 Write a note on scope of synthetic biology